Problem Set 8: Environmental, Resource & Conservation

Microeconomic Theory for Experimentalists & Applied Microeconomists

Due: [**date — before Lecture 1 of Module 9**]

Ground rules: collaboration on ideas is encouraged; write-ups are individual. You may use AI tools for parts flagged with ⟨AI⟩ and must attach the transcript. Show all calculations.

Weights: Part A — 30 points. Part B — 45 points. Part C — 25 points.

Part A: Theory — Guided Calculations (30 points)

Question 1 (The commons, with arithmetic — 15 points)**.** *Four users share a resource pool that starts at 40 units. Each round, each user requests an extraction of 0–10 units; if total requests exceed the pool, it is split proportionally and exhausted. Whatever remains after extraction grows by a factor of 1.5, capped at 40. The game lasts six rounds (all players know this).*

- (a) ***(4 pts)*** *Find the largest total per-round extraction T that keeps the pool at 40 forever, and the corresponding symmetric per-user extraction (round to the nearest feasible integer). Compute each user's total earnings over six rounds on this sustainable path, and compare to the earnings from an all-grab in round one (everyone requests 10).*
- (b) ***(4 pts)*** *The knife: consider a two-round version. Three users extract the sustainable 3 per round throughout. A fourth deviates, requesting 10 in round one and 3 in round two. Compute the deviator's and a cooperator's two-round earnings (track the pool level carefully), and state in one sentence why this makes the situation a social dilemma — note where the deviation's cost appears, and why the full six-round horizon is what completes the dilemma.*
- (c) ***(3 pts)*** *State the standard-theory prediction for this game played by rational money-maximizers without communication, and the prediction for the same game with a 30-second non-binding chat before each round. One sentence each.*
- (d) ***(4 pts)*** *Ostrom, Walker & Gardner (1992) found that face-to-face communication raises cooperation dramatically and durably. In one paragraph: why is this a puzzle for the theory in (c), and name the three candidate mechanisms from lecture (preferences, beliefs, identity) with one sentence on what each claims the talk is doing.*

Question 2 (PES, inframarginality, and the auction — 15 points)**.** *A payment-for-ecosystem-services program pays forest owners \$20 per hectare per year to keep forest standing. It enrolls 1,000 hectares. Without payments, 10% of these hectares would be deforested each year; with payments, 5% are. Each deforested hectare releases 100 tons of CO_2.*

- (a) ***(4 pts)*** *Compute the program's annual cost, the hectares of deforestation averted per year, the tons of CO_2 averted, and the cost per ton averted.*

(b) ***(4 pts)*** *Consider a second program with the same terms but different enrollment: 500 of its 1,000 enrolled hectares belong to owners who would never have deforested (the payments to them are inframarginal); on its at-risk 500 hectares, deforestation falls from 10% to 5%. Compute this program's cost per ton averted, and state in one sentence why asking applicants "did you plan to cut?" cannot fix this.*

(c) ***(4 pts)*** *Explain how a procurement auction (Jack 2013) addresses inframarginality: what do bids reveal, who wins contracts under a budget, and why does self-selection do the targeting that screening questions cannot? Note one honest limitation of the auction (e.g. what happens if owners collude, or if opportunity cost and environmental value are negatively correlated).*

(d) ***(3 pts)*** *Much of the program's carbon benefit arrives near 2100. In three sentences a policymaker would accept: why does the choice between a 1% and a 7% discount rate change the program's present value by roughly two orders of magnitude, why is that choice normative rather than technical, and what should an honest cost-benefit report do about it?*

Part B: Experimental Design (45 points)

Question 3 (A conservation procurement auction, in the field)**.** *Design a field experiment comparing a* procurement auction *for conservation contracts against a* flat-rate *PES program of equal budget, in a smallholder forest landscape. Specify:*

(a) ***(6 pts)*** *Setting and pool: the landscape, the landholders, the conservation contract's terms (what the owner must do, for how long), and why this setting makes the targeting question first-order.*

(b) ***(12 pts)*** *Treatments: (i) flat-rate offer arm, (ii) procurement-auction arm (specify the auction format — sealed bid, discriminatory or uniform price — and justify), and (iii) a pure control arm. Randomize at what level, and why? State precisely which comparisons identify (1) the program effect on deforestation, (2) the targeting gain from the auction (cost per averted hectare, auction vs. flat), and (3) what the bid distribution itself tells you.*

(c) ***(9 pts)*** *Measurement: compliance monitoring (satellite + ground checks), opportunity-cost proxies at baseline, and — non-negotiable in this domain —* ***leakage****: how you detect deforestation displaced to unenrolled plots or neighboring villages, and how the randomization level in (b) supports leakage measurement.*

(d) ***(9 pts)*** *Hypotheses and power: anchor the program effect on Jayachandran et al. (deforestation roughly halved) and state the cluster-randomized sample (villages per arm, landholders per village) needed, noting the design effect from clustering; state which comparison (program effect or auction-vs-flat targeting gain) drives the sample size and why.*

(e) ***(9 pts)*** *The biggest threat to YOUR design and your answer to it. (Candidates: leakage across arm boundaries; auction bids contaminated by anchoring on the flat rate if arms share information; monitoring-induced behavior change.)*

Part C: Silicon Pilot Interpretation ⟨AI⟩ (25 points)

Question 4 (The villagers who never broke a promise)**.** *You run `module8-simulation-lab.ipynb`. With communication, your silicon villagers' extraction drops to almost exactly the sustainable level* *and stays there; the pre-round messages are articulate mutual pledges;* ***every agent honors every pledge in all rounds****, and there is no defection in the known final round. Human groups with communication improve dramatically but imperfectly: pledge-breaking occurs, cooperation erodes under temptation, and known-endgame defection is common. Your lab partner reports: "communication works even better than Ostrom found."*

(a) ***(8 pts)*** *Explain why perfect pledge-keeping and the absence of endgame defection are* divergences *from human behavior rather than a stronger version of Ostrom's result, and give two mechanisms (at least one LLM-specific) that could produce them.*

(b) ***(9 pts)*** *Design the diagnostic battery: your design must include (i) a* temptation spike *(e.g. the final round's extraction value is doubled, and agents are told) and (ii) a* message-ablation *arm (replace the agents' real messages with content-free filler while keeping the communication structure), and may add others (anonymity framing; a mixed group with one profit-focused persona). State the predicted result pattern under each mechanism from (a). Run it if API budget allows and report; otherwise state predictions precisely.*

(c) ***(8 pts)*** *A colleague designing a human CPR experiment with a communication treatment asks what your sim can offer. State one thing it CAN legitimately inform (and why), one thing it CANNOT (and why), and one concrete design change to the human experiment that your silicon results — read correctly — would justify.*

Attach your AI transcripts and, if you ran code, the results CSV.